

Kubernetes and Cloud Native Associate - KCNA

Container Orchestration Security

Security Kubernetes Security Primitives

Hello, and welcome to this lesson on Kubernetes security primitives. In this article, we explore the foundational security measures required for running production-grade applications on Kubernetes. Understanding these security components is essential for maintaining a robust and secure cluster.

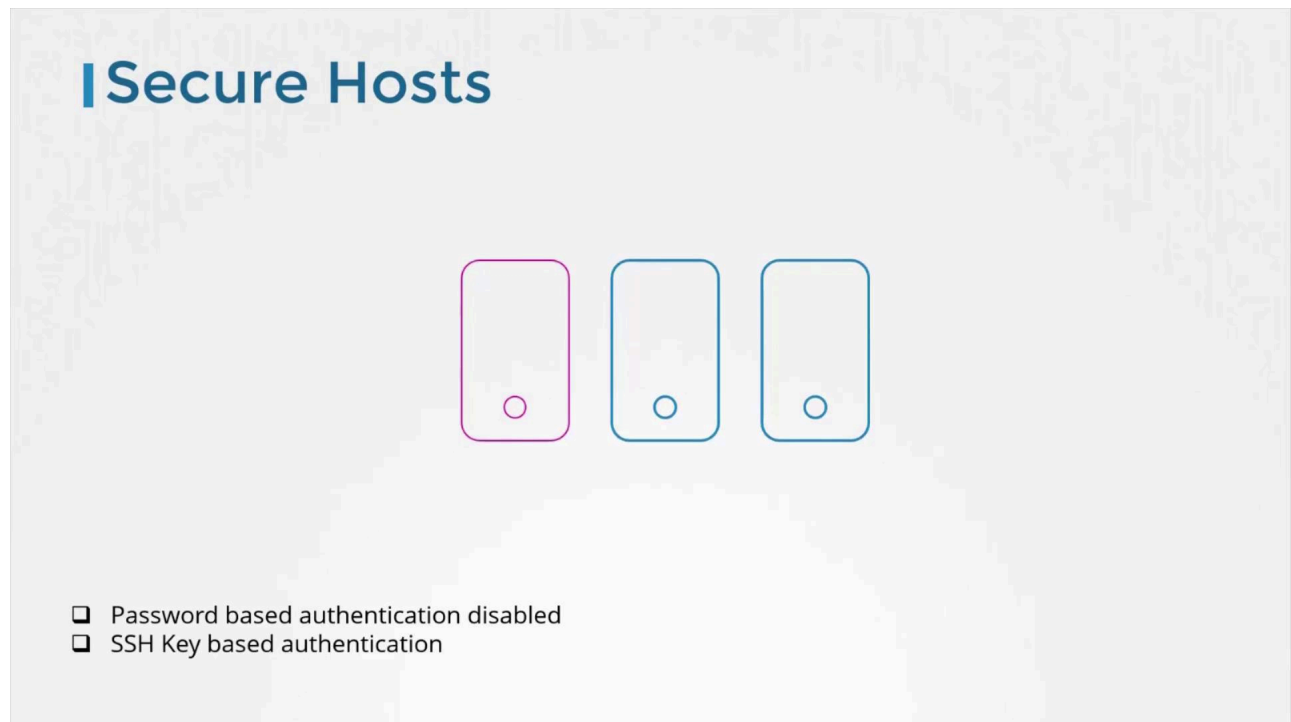


Securing the Underlying Hosts

Before diving into Kubernetes-specific features, it is vital to secure the physical or virtual hosts that form the cluster. Ensure all host access is protected by:

- Disabling root access
- Disabling password-based authentication
- Enforcing SSH key-based authentication
- Implementing other critical security measures

A compromise at this level can jeopardize the entire infrastructure.



Kubernetes-Specific Security Measures

Focusing specifically on Kubernetes, the core component is the kube-apiserver. This component is the central point of contact for operations, whether accessed via the kubectl utility or directly through its API. Its security is paramount, so consider these key questions:

- Who can access the cluster?
- What actions can they perform once access is granted?



Authentication

Access to the API server is managed through robust authentication mechanisms. Methods include:

- Static files with user IDs and passwords
- Tokens and certificates
- External authentication providers (e.g., LDAP)
- Machine-to-machine interactions managed through service accounts

| Authentication

Who can access?

- ☐ Files – Username and Passwords
- ☐ Files – Username and Tokens
- ☐ Certificates
- ☐ External Authentication providers - LDAP
- ☐ Service Accounts

Authorization

Once authenticated, the system determines valid actions through authorization. Kubernetes primarily uses Role-Based Access Control (RBAC), where permissions are assigned based on roles or group memberships. Additional methods include Attribute-Based Access Control (ABAC) and webhooks for extended flexibility.

| Authorization

What can they do?

- ☐ RBAC Authorization
- ☐ ABAC Authorization



Secure Communication Within the Cluster

All communications between essential cluster components—such as the etcd cluster, kube-controller-manager, scheduler, API server, and nodes (kubelet and kube-proxy)—are secured with TLS encryption. This ensures that every interaction within the cluster remains confidential and tamper-proof.

TLS Encryption

An entire section of Kubernetes documentation is dedicated to configuring TLS certificates. These certificates secure intra-cluster communication, safeguarding sensitive data exchanged between:

- Kube ApiServer
- ETCD Cluster
- Kubelet
- Additional components

TLS Certificates

ETCD



Beta



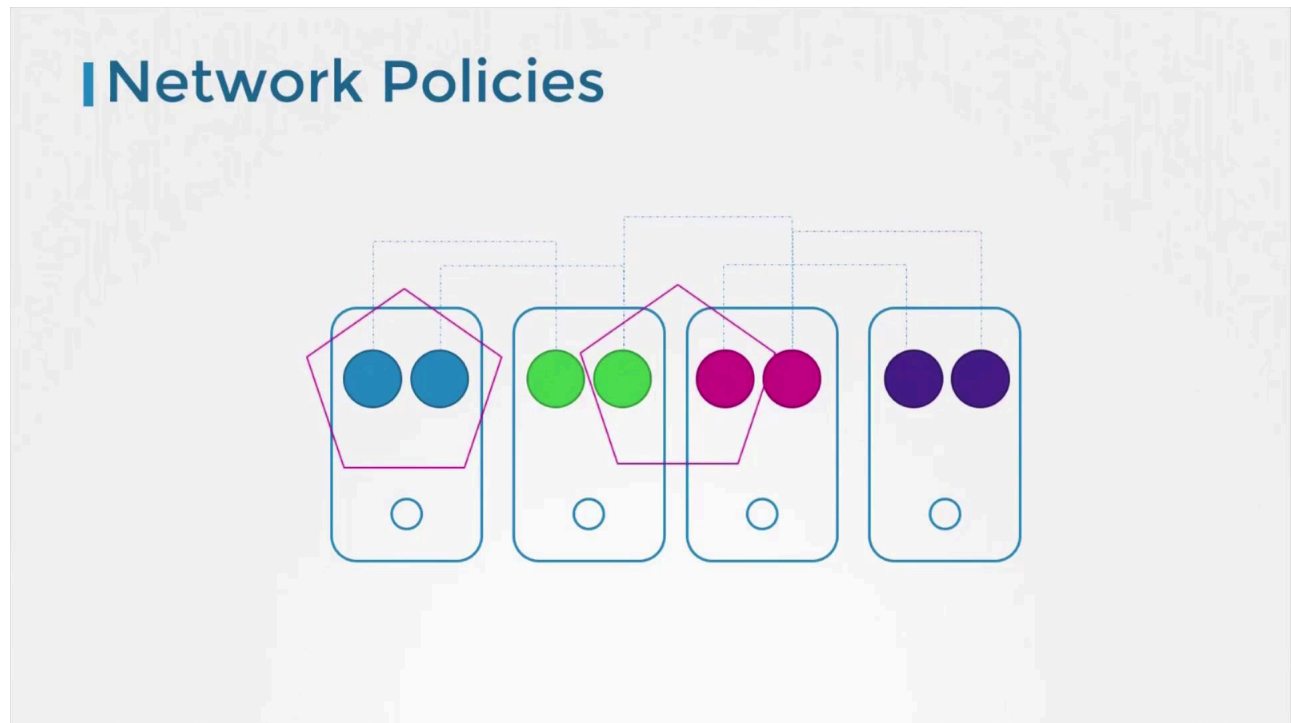
● Search docs...



Kubernetes and Cloud Native Associate - KCNA ▾

Network Policies

By default, pods within the cluster can communicate freely. However, to tighten security, you can implement network policies, which restrict communication between pods and enhance overall cluster security.



This high-level overview has introduced the key security primitives in Kubernetes. In the upcoming sections, we will explore each component in greater depth to help you build a secure and resilient Kubernetes cluster.

For additional insights, consider exploring the following resources:

- [Kubernetes Documentation](#)
- [Kubernetes Basics](#)
- [Docker Hub](#)
- [Terraform Registry](#)



Watch Video

Watch video content

Previous

◀ **Configuring Kubernetes Scheduler Profiles**

Next

Authentication ▶